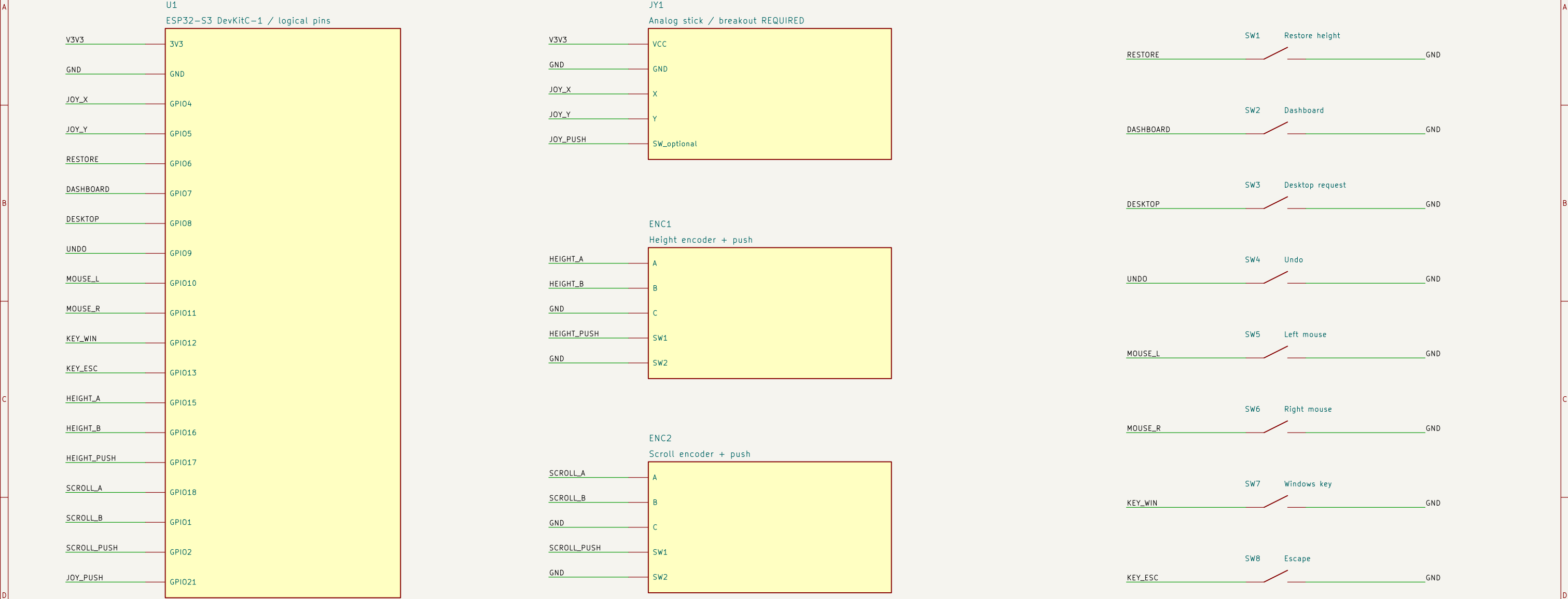


RIG INPUTS / HAND-WIRED PROTOTYPE A

8 MX keys + 2 push encoders + analog mouse stick / one native USB cable



CONNECTIONS: identical net labels are electrically connected. Pin names are functional markings, NOT physical header positions.

USB: PC -> board native USB/OTG socket. GPIO19 and GPIO20 reserved. UART-only USB socket cannot carry our HID reports.

KEYS: two electrical contacts per MX switch; either orientation. Enable INPUT_PULLUP on all keys, encoder A/B and push inputs.

ENCODERS: passive mechanical A/B/C + isolated push switch. Join C and SW2 locally to GND; 4 wires per encoder module.

JOYSTICK: 3.3 V analog assumption ONLY. Flex pin count/order/pitch and click are UNVERIFIED. No physical FPC pin assignment.

Four-wire stick: leave GPIO21 unconnected, disable its action; SW5 is left click. Five-wire breakout: click may also use GPIO21.

No matrix, diodes or external pull-ups in minimum build. Firmware debounce required. Optional 100 nF from each ADC input to GND.

Use 2.54 mm solder headers or prewired keyed plugs. Keep GND with each cable. Insulate solder joints and add strain relief.

REFERENCE BOARD ONLY: check exact clone schematic before wiring. No PCB footprint or manufacturing pinout is implied here.

Logical wiring; joystick flex mapping pending		
Sheet: / File: rig-inputs.kicad_sch		
Title: Rig inputs – hand-wired prototype		
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